

MICS Plus Tabulation

A step-by-step guide to using the application

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Start here

This guide helps you turn the files supplied by the MICS team into Excel output tables. You do not need to write calculations or change computer code. Follow one step at a time.

Your route from setup to finished tables

1. **Set up your computer** - install R and an editor such as RStudio.
2. **Create or open a survey project** - choose where your work belongs.
3. **Install the package and start the app** - follow sections **1D** and **1E**, then work through these tabs inside the app:
 - a. **Download Files** - download tabulation plans and preparation files to your project.
 - b. **Data Preparation** - select your survey documents and authorized data, then click **Set**.
 - c. **Tabulator**, then **Consistency Checks** - calculate and review a table.
 - d. **Write to Excel** - save the reviewed table.

The later pages explain running several tables, downloading data, and optional viewing tools. You can leave those until you have completed your first table.

Words used in this guide

- **Click:** press the left mouse button once. On a touchpad, tap once.
- **Tab:** a section name across the top of the application.
- **Browse:** open a window to find a file on your computer.
- **Folder:** a place that holds files. A preparation folder may contain smaller folders.
- **Project:** a folder for one piece of survey work, with a file that helps RStudio reopen it.
- **Console:** the panel where you paste R commands and see messages.
- **Package:** an add-on for R; `micsPlusTableR` contains this application.
- **Sheet:** one page of an Excel workbook, such as **1.1a**.
- **Wave:** one round of the survey. Use the wave your team gave you.

About the pictures. The Download Files picture shows the new tab in the 0.3.0 development version, before loading the survey list. Other pictures are actual screenshots from version 0.2.7, using Mongolia Wave 1 as an example. Your file names and results will differ. Use the files for your own survey. Screenshots show a temporary example output folder; your project normally saves to **micsPlusTableR-output**. In the HTML guide, click a picture to enlarge it; press **Esc** to return.

Looking for package functions? Use the [package reference manual](#) and [package documentation](#). Those documents cover programming and technical details.

1. Set up your project and open the application

1A. Choose where to run R

If you are new to R, follow the **RStudio Desktop steps below**. R does the calculations; RStudio gives you a window in which to start them. These are two separate programs. The tabulation application then runs in your web browser. You only need to copy a few commands; the table-making steps use buttons.

An **IDE** is a program that brings together tools for working with code. The package can run in an R session started from different editors and code systems. You only need one of these options:

Option	What you need to know
RStudio Desktop	The route explained here, with an R Console and a project menu. Install R separately.
Positron	An alternative with built-in R support. Install R, open your survey folder, and choose an R session.
Visual Studio Code (VS Code)	Install R and configure the R extension; run the commands in an R terminal.
Cursor	Use its code editor with R installed and a compatible R extension, or start R in its terminal.
JupyterLab / Jupyter Notebook	Use an R kernel, such as IRkernel. This requires extra setup.
R's own console / a terminal	Run the same R commands from your survey folder. No separate IDE is required.

For setup details, see [Positron's R guide](#), [VS Code's R guide](#), [Cursor's extension guide](#), and [IRkernel for Jupyter](#). The optional [other environments](#) section explains project folders and browser access for these routes. The remaining click-by-click computer setup instructions use RStudio Desktop.

What belongs to you: create or use your own survey project folder and install the **micsPlusTableR** package. Your project holds your survey work and generated results.

1B. Install R, then RStudio Desktop (once)

If both programs already open on your computer, go to **1C**. You need internet access for installation. Use a recent R version supported by your editor; the package requires **R 4.2 or later**. You also need a web browser and an Excel viewer to review the saved workbooks. The workbook preview may also need internet when the app opens; see [Internet access and the workbook preview](#) for the role of browser caching.

Install R first

1. Open your web browser and go to the [official R download site](#).
2. **Windows:** click **Download R for Windows**, then **base**, then the download link for R. Open the downloaded **.exe** file from **Downloads**. Follow the installer, using its standard options, and click **Finish**.
3. **Mac:** click **Download R for macOS**. Choose the **.pkg** installer matching your Mac and macOS version. **Apple menu > About This Mac** shows whether your Mac has an Apple chip or an Intel processor. Open the downloaded file and follow the installer.
4. **Linux:** follow the R download page's instructions for your Linux distribution, with your support person's help if needed.

Install RStudio Desktop next

1. Go to [Posit's RStudio Desktop download page](#). Choose the **free RStudio Desktop** download for your operating system.
2. **Windows:** open the downloaded installer and follow its steps. **Mac:** open the downloaded **.dmg** and drag RStudio into **Applications**. **Linux:** use the installer and instructions for your distribution.
3. Open **RStudio** from the Windows **Start** menu or Mac **Applications**. If it asks you to choose R, select the version you just installed.
4. Find the **Console** panel. When you see a **>** prompt, R is ready.

Ready to continue: RStudio opens and its Console shows **>**. If you see "R not found," finish installing R, close RStudio, and reopen it. If your computer requires an administrator password you do not have, ask your IT support person to complete the installation.

1C. Create or open your survey project

Do this before starting the application. Choose one route below. A project keeps each survey's files and results together. Use a memorable name, such as **Mongolia-Wave1**; substitute your own survey.

New work: create a new project folder

1. In RStudio, click **File > New Project**.
2. Choose **New Directory**, then **New Project**.
3. Under **Directory name**, enter your survey project name.
4. Beside **Create project as subdirectory of**, click **Browse** and choose a location you can find again, such as **Documents**.
5. Click **Create Project**. RStudio creates and opens the folder.

You already have a folder, but no .Rproj file

1. Click **File > New Project > Existing Directory**.
2. Click **Browse** and choose your existing survey folder.
3. Click **Create Project**. RStudio adds a project file to that folder; your existing survey files stay in place.

You already have an RStudio project

1. Click **File > Open Project**.
2. Find your survey folder, select its **.Rproj** file, and click **Open**. You can also double-click that file in File Explorer or Finder.

These routes follow the [RStudio project instructions](#). To switch surveys later, stop the app and use **File > Open Project** again.

Check your location: the project name should appear at the top right of RStudio.

Your files: keep your supplied survey data, Excel plan, and complete preparation folder inside an **inputs** subfolder if convenient. Copying files into the project does not load them into the app; you will select them with **Browse** in sections 2 and 3. Files may also remain in another accessible folder. Keep the project available while working; extract ZIP folders first if you downloaded the inputs as ZIP folders.

The app normally creates **micsPlusTableR-output** inside your project for Excel results. Browser downloads may go to **Downloads** instead. The **Write to Excel** screen shows the actual saved workbook paths.

1D. Install the micsPlusTableR package (once)

1. Keep your survey project open. Click the **Console** beside **>**. Use the **Console**, rather than the **Terminal** tab or a search box.
2. Install **remotes** only if it is not already available in your R library. Once installed, you do not need to install it again: skip the first line on later uses. Run the second line to install **micsPlusTableR**. Paste each needed command into the Console and press **Enter** (**Return** on a Mac). Wait for **>** to return before running the next command:

```
install.packages("remotes")
remotes::install_github("barisr/micsPlusTableR")
```

Copy only the command, without the surrounding box or any **>** prompt. On Windows, use **Ctrl+C** to copy and **Ctrl+V** to paste; on a Mac, use **Command+C** and **Command+V**. **remotes** is the helper that installs the application package from GitHub. The second command installs **micsPlusTableR** and its required packages, including those used by the supported survey preparation scripts. This may take several minutes. Leave RStudio open while messages scroll.

3. If R asks to create a personal library, accept it. If asked to choose a CRAN download mirror, choose **0-Cloud** if offered. If asked which existing packages to update, choose the numbered **None** option unless your support person has requested updates. Required missing dependencies still install.
4. If an error appears, copy the message and ask your support person for help. A returned **>** alone does not prove installation succeeded. For folder or lock errors, see [Installation cannot finish](#).
5. Click **Session > Restart R**, then run this check in the Console:

```
packageVersion("micsPlusTableR")
```

Ready to continue: a version number appears without an error saying the package is missing. Continue to **1E** to launch the application.

You normally install once for the R library you use, not once per survey or on every launch. A different R installation or an isolated project library may need its own package installation. To update, finish and save your work, stop the app, and follow your survey lead's instructions. Run the GitHub installation command again and restart R before using the updated package.

1E. Start the application (each time you work)

1. Open your survey's **.Rproj** file. Confirm its name at the top right of RStudio. You do not need to repeat installation for ordinary launches.
2. In the **Console**, paste this line and press **Enter**:

```
micsPlusTableR::run_app()
```

3. The application opens in a browser on **Data Preparation**. On a narrow window, use **Toggle navigation** to see the tabs. If you need plans or preparation files, open **Download Files**.
4. Follow **section 2** to locate your household data, household members data, Excel tabulation plan, and preparation folder. Then follow **section 3** to select the country and wave, upload the files, and click **Set**.

You are ready when: you can open **Download Files**, **Data Preparation**, **Tabulator**, **Consistency Checks**, and **Write to Excel** across the top. Click **User's Guide** to read these instructions inside the application. Use **Open full guide** there if you want a separate browser tab.

While working: keep RStudio (or your chosen R environment) and the application open. A busy Console without a > prompt is normal while the app runs. Do not paste more commands there. Do not refresh the browser; you may lose work not yet written to Excel.

No browser window? Find the address beginning **http://127.0.0.1:** in the Console. Copy the complete address into your browser's address bar and press **Enter**. The number at the end can change each time you start.

Where the work happens: with this local setup, the browser connects to R on your computer. Selecting input files loads them into the running app; it does not install them into the package. Opening a project later does not restore the app's uploaded inputs or unsaved results.

For help, show your support person the Console message and the [package setup instructions in Section 1D of this guide](#).

Internet access and the workbook preview

Why the preview uses a browser library. The workbook preview in **Data Preparation** shows the uploaded Excel tabulation plan inside the app. A browser cannot display an Excel workbook as a table on its own, so the app uses a JavaScript library called **SheetJS** to read it and convert the selected sheet into a displayable table. You do not need to install this library yourself. It is separate from the calculations performed by R and the calculated table preview in **Tabulator**.

When internet is needed. When the app opens, the browser loads SheetJS from **cdn.jsdelivr.net** if it cannot reuse a saved copy. In a local RStudio session, the workbook itself is read from the running app on your computer; the preview code does not send it to that external server. Once the library has loaded, changing the preview sheet does not require downloading the library again while that app page stays open.

What caching means for later use. The browser may keep a copy of the library in its cache and reuse it on later visits, avoiding another download. However, caching is **not a guarantee of offline access**: the copy can be cleared or expire, and another browser or a private window may not have it. Browser settings can also require a new request. See [how browser caching works](#).

Before working offline: open the app while connected, upload the plan, and check that its workbook preview appears. Keep that app page and R session open. Calculations and Excel output do not depend on SheetJS; with the required packages and input files available locally, they do not need this external download. Installation, **Download Files**, and retrieving cloud-only inputs still need internet. If only the workbook preview fails, you can inspect the plan in Excel and continue tabulating.

2. Gather your survey files

You need two sets of files for the **same country, period, and wave**:

- **Household and household-member data (.sav):** obtain them from the [MICS surveys page](#) with the required permission. These data are never distributed through this files repository or the package's download functions.
- **Tabulation plans and preparation files:** download them from [micsPlusTableR-Files](#) using the app or one of the alternatives below. These downloads require no GitHub account once the repository is public. Before publication, ask your survey lead for them.

Download with the app

1. Open **Download Files** and click **Load / refresh surveys**. Internet access and a public files repository are needed to load the list.
2. To download **one** folder, keep **Selected surveys** and tick that survey, such as JAM_W1. To download **several**, tick each survey you want. For **every** published survey folder, select **All surveys**.
3. Review **Selected survey folders** for the file count and total size.
4. Check **Download folder**. It starts at the current project you opened before launching the app. For example, selecting JAM_W1 saves its materials in a JAM_W1 subfolder directly inside that project.
5. To choose another destination, click **Browse folders...**, select a folder, and click **Select**, or type a path. You may create a folder in the picker. A relative path such as **inputs** means a subfolder of the current project. **Use current project** restores the original default.
6. Leave **Replace existing files** off to protect local copies. If the selected files already exist, use a different folder or turn it on deliberately.
7. Click **Download files** and wait for the success message, which shows where the files were saved. The preparation subfolders are preserved.
8. Open **Data Preparation**. Select the downloaded plan and complete preparation folder, along with the household and member data you obtained separately from the [MICS surveys page](#) with permission.

Files are saved directly by R to the chosen location. They do not go through your browser's **Downloads** folder. In a remotely hosted R session, the path is on that remote machine; use your host's file access tools to retrieve files.

Download Files screen

The screenshot shows the 'Download Files' screen, which is divided into two main sections: a left sidebar for controls and a right main area for instructions and status.

Left Sidebar Controls:

- A button labeled 'Load / refresh surveys'.
- Section: 'Which survey folders?'
 - ☒ Selected surveys
 - ☐ All surveys
- Text: 'Choose one or more surveys'
- Section: 'Download folder'
 - Text input field containing: '/private/tmp/mics-download-tab/survey-pro'
 - Buttons: 'Browse folders...' and 'Use current project'
- Text: 'Each survey gets its own subfolder in this location.'
- ☐ Replace existing files
- A large dark blue button labeled 'Download files'.

Right Main Area:

- Section: 'Download tabulation plans and preparation files'
 - Text: 'Load the available surveys, choose one or more folders (or all surveys), then choose where to save them.'
 - Text: 'Household and household-member data are obtained separately from the [MICS Plus website](#) with the required permission. They are not included in these downloads.'
 - Text: 'Current project: /private/tmp/mics-download-tab/survey-project'
- Section: 'Selected survey folders'
 - Text: 'No survey list loaded yet.'
 - A blue button with text: 'Click Load / refresh surveys to see the published materials.'
- Section: 'After downloading'
 - Text: 'Open Data Preparation and select the downloaded Excel plan and complete preparation folder, together with your separately obtained household and household-member data.'
 - Text: '[Alternative: download all plans and preparation files as a ZIP](#)'

Figure 1: Download Files: survey selection and destination controls are on the left; instructions and download status are on the right.

This picture shows the screen before **Load / refresh surveys** is clicked. Your own project path appears in **Download folder**. After loading, the available survey checkboxes appear under **Choose one or more surveys**.

Alternative: download a ZIP through your browser

1. Click [Download all plans and preparation files \(ZIP\)](#).
2. Find the ZIP in **Downloads**. On Windows, right-click it and choose **Extract All**; on a Mac, double-click it.
3. Open the extracted micsPlusTableR-Files-main folder. Copy the survey folder you need, such as JAM_W1, into your survey project's **inputs** folder. Keep its preparation folder and all subfolders together.

Alternative: download from the R Console

Open your survey project. Run these commands in the **R Console before starting the app**. If the app is running, stop it first so the `>` prompt returns. First list the available surveys:

```
available <- micsPlusTableR::list_survey_files()
unique(available$survey)
```

Choose **one** of these alternatives:

```
# One survey
micsPlusTableR::download_survey_files("JAM_W1")

# Several surveys
micsPlusTableR::download_survey_files(c("JAM_W1", "MNG_W2"))

# All published surveys
micsPlusTableR::download_survey_files()
```

The Console function defaults to **inputs** under `getwd()`, while the app tab defaults to the project itself. Console downloads preserve the survey and preparation subfolders. Use `dest_dir = "other-folder"` to choose another destination. Existing files are protected: use a new folder, or add `overwrite = TRUE` only when you intend to replace them.

For individual files, inspect `available$path` and pass the exact paths:

```
micsPlusTableR::download_survey_files(
  files = available$path[1],
  dest_dir = "selected-inputs"
)
```

Select a whole survey folder to include its preparation helpers. Add the separately authorized household/member data before running the app. If GitHub limits requests, retry later or use the ZIP link.

Check your input set

Keep these **four items for the same country, period, and wave** together in your survey project's **inputs** folder, or another folder you can find again. The household and member files come from the MICS Plus website with permission; the Excel plan and preparation folder come from `micsPlusTableR-Files`. Once you have both, run `micsPlusTableR::run_app()`.

Item to obtain	How to recognize it	Where it goes in the app
Household data	Name often ends in <code>hh.sav</code>	Select household file
Household members data	Name often ends in <code>hl.sav</code>	Select household members file
Tabulation plan	Excel file ending in <code>.xlsx</code>	Select tabulation file
Preparation folder	Folder containing the team's preparation file	Select preparation folder

Your team may supply longer names. Do not rename the files just to match the pictures. You do not need to open the `.sav` files or the preparation code.

How to choose a file

1. Click **Browse...** beside the relevant item in the application.
2. In the window that opens, find the folder your team gave you. If it was downloaded, first look in **Downloads**.
3. Double-click a folder to open it. Click the required file once to select it.
4. Click **Open**. Wait until its name appears in the application and the upload finishes. Repeat for the other files.

How to choose the preparation folder

1. If the team supplied a ZIP file, extract it first: on Windows, right-click it and choose **Extract All**; on a Mac, double-click it.
2. Beside **Select preparation folder**, click **Browse...**
3. Select the folder that contains the preparation file, then click **Upload**, **Select**, or **Open**, depending on your browser. Select the folder itself, rather than an individual file inside it.
4. If the browser asks to upload the folder's files, check its name and confirm.

Check before continuing: use the complete preparation folder supplied by your team. Keep any folders inside it, such as **FIES-inputs**, in place. If you cannot tell which files belong together, ask your survey lead.

3. Choose the files and click Set

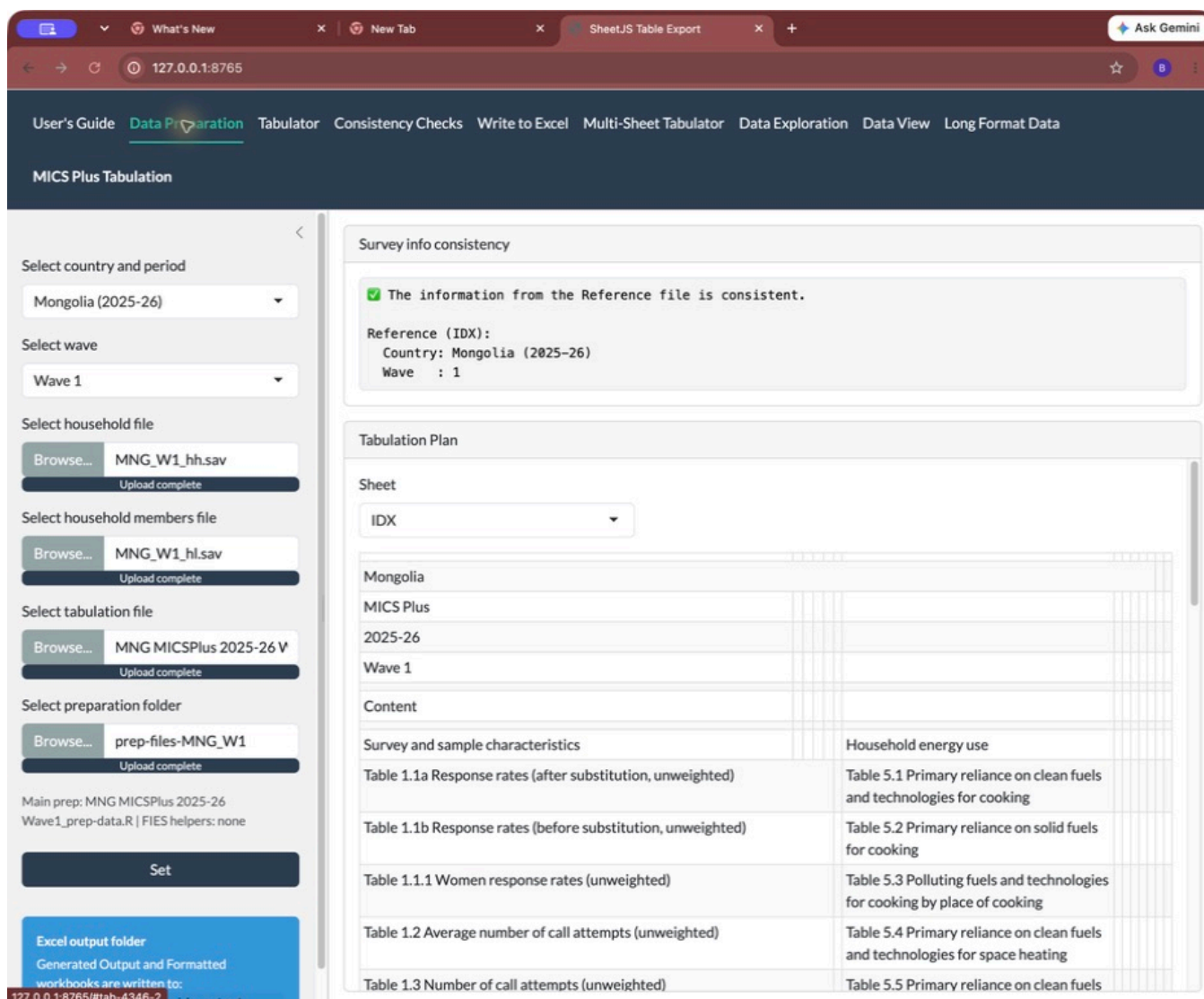


Figure 2: Data Preparation: survey choices and four Browse controls are on the left; the survey match message and Excel plan are on the right.

1. Click **Data Preparation**. Under **Select country and period**, click the small arrow and choose your survey. Choose the matching **wave** below it.
2. Use the four **Browse...** controls as described on the previous page.
3. Read **Main prep** below the folder control. Check that it names your team's preparation file. **FIES helpers: none** is normal for surveys without FIES.
4. Click **Set** once. If it is below the visible area, scroll down inside the left panel. Wait for preparation to finish.
5. Read **Survey info consistency** at the upper right. Check the country, period, and wave. The example says the reference information is consistent.

Ready to continue: the survey details agree and there is no preparation error. A matching survey message alone does not resolve a separate error. If a mismatch or error appears, check your choices or ask your survey lead.

4. Calculate and look at one table

The screenshot shows the MICS Plus Tabulation web application. On the left, the 'Select a sheet' dropdown is set to '1.1a'. The 'Format type' is set to 'Formatted' (radio button selected), and the 'Pivot type' is set to 'Header' (radio button selected). A 'Run' button is visible. The main area displays a 'Pivot table preview' for 'Table 1.1a Response rates (after substitution, unweighted)'. The table has columns for Area (Total, Urban, Rural), Region (Western, Khangai, Central, Eastern, Ulaanbaatar), and Degree of urbanization (City, Town & Semi-dense area, Rural area). Rows include various household sampling and response status metrics.

	Area			Region					Degree of urbanization		
	Total	Urban	Rural	Western	Khangai	Central	Eastern	Ulaanbaatar	City	Town & Semi-dense area	Rural area
(Households sampled) Households sampled	3,000	1,557	1,443	545	547	594	595	719	623	930	1,447
(Households sampled) Interviewed	2,681	1,387	1,294	499	498	529	522	633	539	838	1,304
(Households sampled) No consent for next wave	97	68	29	10	14	18	17	38	37	35	25
(Households sampled) With consent for next wave	2,584	1,319	1,265	489	484	511	505	595	502	803	1,279
(Households sampled) Refused	73	52	21	4	9	14	14	32	28	24	21
(Households sampled) No consent for next wave	63	45	18	3	9	12	13	26	24	21	18
(Households sampled) With consent for next wave	10	7	3	1	0	2	1	6	4	3	3
(Households sampled) No eligible respondent	2	1	1	1	0	1	0	0	0	1	1
(Households sampled) Phone number(s) does not belong to sampled household	33	21	12	4	5	6	6	12	13	7	13
(Households sampled) Phone number(s) inactive	43	19	24	7	3	11	14	8	5	13	25

Figure 3: Tabulator: select a sheet on the left and click Run; the calculated table appears on the right.

1. Click **Tabulator** at the top.
2. Under **Select a sheet**, click the arrow. Choose the table code requested by your survey lead. The picture uses **1.1a**.
3. Leave **Format type** on **Formatted** and **Pivot type** on **Header** for an easy-to-read preview.
4. Click **Run** once and wait for the table to appear.
5. Read **Table name** above the result. Check that it is the table you intended to produce. Scroll inside the preview to see more rows or columns.

Ready to continue: the correct table name and calculated values appear. The next step is **Consistency Checks**. A preview has not yet saved your calculated table to an Excel file.

If your reviewer asks for plain numbers, choose **Unformatted**. Choose **Suppressed** to inspect the small-sample display rules. Keep **Header** selected unless your reviewer asks for another view.

4A. Understand worksheet filters

This section is for people who prepare or review the Excel plan. It explains which records a `filter(...)` instruction includes when the table is calculated. These are worksheet instructions, separate from the app's search boxes.

Find the filter instructions. The plan's condition row contains the `.sav` data source in column B, on row 4, 5, 6, or 7. Filters for columns appear on that same row. The statistic written in the table body is called `stat_type`: for example, `p` calculates a percentage, `n` a weighted count, and `n_unw` an unweighted count.

Column B is global. A filter such as `filter(age >= 18)` in B limits the whole table to adults. It remains active when another filter starts, when the statistic changes, and when a new table row begins. A later filter cannot bring excluded records back into the table.

Additional filters apply to horizontal blocks. In a horizontal table, a `filter(...)` in C or a later column combines with B. It applies to its own column and extends to the right. Its position relative to B makes no difference: it always combines with B. Changes from `p` to `n` or `n_unw` keep the filter, so percentages and their counts use the same population.

There are two ways an additional filter stops:

- **A non-mean statistic changes to a mean.** For example, when `n` is followed by `mean(age)`, that mean cell and following cells use B alone. Returning to `n` does not restart the old filter. This rule is checked across columns separately for each row; changes down rows do not stop it.
- **Another explicit filter starts.** The new filter replaces the previous additional filter and combines with B. It applies at its own starting column, including when that column calculates a mean.

Changes among non-mean statistics, between mean variables, and from a mean to a count do **not** stop the filter. A filter starting on a mean continues through later counts; a subsequent change from a non-mean to a mean stops it unless another explicit filter starts there.

Blank filter cells do not cancel an active filter. To remove an additional restriction explicitly, write `filter(TRUE)`; B still applies. To keep an additional restriction when entering a mean from a non-mean, repeat its `filter(...)` above that mean column.

How to recognize the table direction. The reader selects horizontal mode when the plan has `n_unw` immediately before `IDX`, or a `100/100.0` statistic. Other plans use vertical mode. In vertical tables, the first filter in B still restricts the common dataset; additional horizontal filter blocks are not processed. Check with the plan author before adding such blocks to a vertical table.

Filter example: follow one row from left to right

Suppose B contains `filter(age >= 18)`. In one horizontal table row, the instructions are:

Column	Statistic	Additional filter starts here	Records available to this cell
C	p	None	All adults
D	p	<code>filter(sex == 2)</code>	Adults with sex code 2
E	n	None	Adults with sex code 2
F	n_unw	None	Adults with sex code 2
G	mean(age)	None	All adults: entered a mean from a non-mean
H	n	None	All adults: the old filter stays off
I	mean(income)	<code>filter(region == 1)</code>	Adults in region 1
J	mean(age)	None	Adults in region 1
K	n_unw	None	Adults in region 1

No cell includes people under 18. D's filter applies to D-F, including the percentage and its counts. It stops at G. I starts a new filter, which applies to I-K and does not include D's sex restriction. Other row and column conditions still determine what is measured within these populations.

Which changes count? The engine uses the effective statistic after the worksheet reader fills missing statistic cells and ignores surrounding spaces when identifying means. Supported mean forms are `mean(variable)`, `mean`, and the legacy `Mean` form. A blank statistic cell does not itself mean "stop the filter." Changes such as p to n, n to n_unw, `mean(age)` to `mean(income)`, and `mean(age)` to n all preserve it. Only entering a mean from a non-mean stops an inherited additional filter.

When filters run. For each required population, the engine loads the selected source, applies B, applies any active additional filter, and then runs the block's calculations. It calculates cell values from that data. The filters therefore affect both the numerator and denominator of a percentage. A filter variable must already exist before those calculations.

Only filter inheritance stops on entering a mean from a non-mean. Existing data-source, weight, calculation, and row/column-condition instructions keep their roles. A calculation-only instruction does not end the filter's column range.

Filters do not reach left. Adding a filter in G does not change F's calculated mean. Display suppression still needs an unweighted count: if a new filter separates a cell from its count, the engine uses the next n_unw to its right on the same row and data source when no count is already linked. Existing count links and the small-sample thresholds remain in effect.

Review after upgrading. The earlier rule stopped additional filters at every statistic change, including p to n; this could give percentages and their counts different populations. Recalculate affected tables and review denominators and suppression. B remains global, including in later blocks; only non-mean-to-mean transitions now stop inherited filters.

4B. Choose decimal places in the worksheet

This section is for people who prepare or review the Excel plan. Add **d=** inside a statistic's parentheses to choose how many decimal places are shown. This is available for every supported statistic type. If the statistic already has an argument, add a comma before **d=**.

What you want	What to write in the statistic cell	Example display
Whole-number average expenses in table 6.2a	<code>mean(HCS8, d=0)</code>	442,572
Average expenses with two decimals	<code>mean(HCS8, d=2)</code>	442,571.91
Mean indicator with one decimal	<code>mean(d=1)</code>	73.3
Percentage with two decimals	<code>p(d=2)</code>	73.30
Count with one decimal	<code>n(d=1)</code>	881.0
Unweighted count with no decimals	<code>n_unw(d=0)</code>	622
Median with two decimals	<code>median(age, d=2)</code>	35.00
Total/distribution statistic with two decimals	<code>p(100, d=2)</code>	100.00

These numbers illustrate the display, not survey results. The same rule works with other supported statistics, including `mean_unw`, legacy `Mean`, and `p_sum`. A constant can be written as `100(d=2)`. Keep the statistic appropriate to the table's direction and intended calculation.

If you leave out d, nothing changes: the app keeps its established formatting for that statistic and output mode. Means default to one decimal place. For example, `mean(HCS8)` still displays 442,571.9.

Use a whole number of decimal places. `d=0` removes the decimal point; `d=1`, `d=2`, and `d=3` show one, two, and three decimal places, including trailing zeros. Spaces are allowed, so `mean(HCS8, d=0)` is valid. Give `d` only once, as a non-negative whole number, without quotation marks or a calculation. For example, `d=-1`, `d=1.5`, and `d="2"` are invalid.

Where it applies. The Formatted and Suppressed Tabulator views use the requested decimals for visible numbers. The Unformatted view retains the full numeric estimate. Both Excel output workbooks use the requested number format, including workbooks written by Multi-sheet Tabulator and values supplied by preparation scripts. Thousands separators remain in the Formatted preview and Excel. Long Format Data records the setting in `DECIMALS` and display values; `OBS_VALUE` retains full precision.

What stays the same. Display rounding does not change the calculated number, weights, filters, denominators, or consistency checks. Excel keeps numeric estimates at full precision when you specify `d`. Parenthesized estimates follow the requested precision, such as (442,572) for `d=0`; (*) and - remain suppression markers.

Changing only `d` never starts or stops a filter. A column using `n_unw(d=2)` still acts as an unweighted count for direction and suppression. Where the reader fills a blank statistic cell, it carries the decimal setting with that statistic. Save the plan and reload it in the app, then rerun the affected tables to apply your edits.

5. Check the table

The screenshot shows the MICS Plus Tabulation web application. The top navigation bar includes links like 'User's Guide', 'Data Preparation', 'Tabulator', 'Consistency Checks' (highlighted), 'Write to Excel', 'Multi-Sheet Tabulator', 'Data Exploration', 'Data View', and 'Long Format Data'. The main interface has a sidebar on the left with a 'Show only inconsistent cells' checkbox and a 'Run Checks' button. The main content area displays a table of results for 'Table 1.1a Response rates (after substitution, unweighted)'. The table has columns: var_name_row, cond, col_index, stat_type, group_total, total_value, and diff_value. The table shows 12 rows of data, all with a diff_value of 0, indicating consistency.

var_name_row	cond	col_index	stat_type	group_total	total_value	diff_value
HH17	1	3	n_unw	3000	3000	0
HH17	1	4	n_unw	1557	1557	0
HH17	1	5	n_unw	1443	1443	0
HH17	1	7	n_unw	545	545	0
HH17	1	8	n_unw	547	547	0
HH17	1	9	n_unw	594	594	0
HH17	1	10	n_unw	595	595	0
HH17	1	11	n_unw	719	719	0
HH17	1	13	n_unw	623	623	0
HH17	1	14	n_unw	930	930	0
HH17	1	15	n_unw	1447	1447	0

Figure 4: Consistency Checks: click Run Checks and examine the smaller check tabs and their result tables.

1. Click **Consistency Checks**. Confirm the **Table name** matches the table you just ran.
2. Click **Run Checks** and wait for the results.
3. Click each of the smaller check tabs, such as **Row Group (N)**. They compare counts or percentages in different parts of the table.
4. If a tab is red, tick **Show only inconsistent cells** to find the differences. Record the table code and ask your survey lead to review them.

Green: this check ran and found no difference beyond its tolerance. **Red:** a difference needs review. **No color or no rows:** the check may not apply; it does not mean it passed. Read the result as well as the color. The example's **diff_value** column shows zero where the compared totals agree.

Ready to continue: the applicable checks pass, or the survey lead has reviewed and accepted any exceptions. Passing checks do not replace the survey lead's review of the table and its meaning.

6. Save the reviewed table to Excel

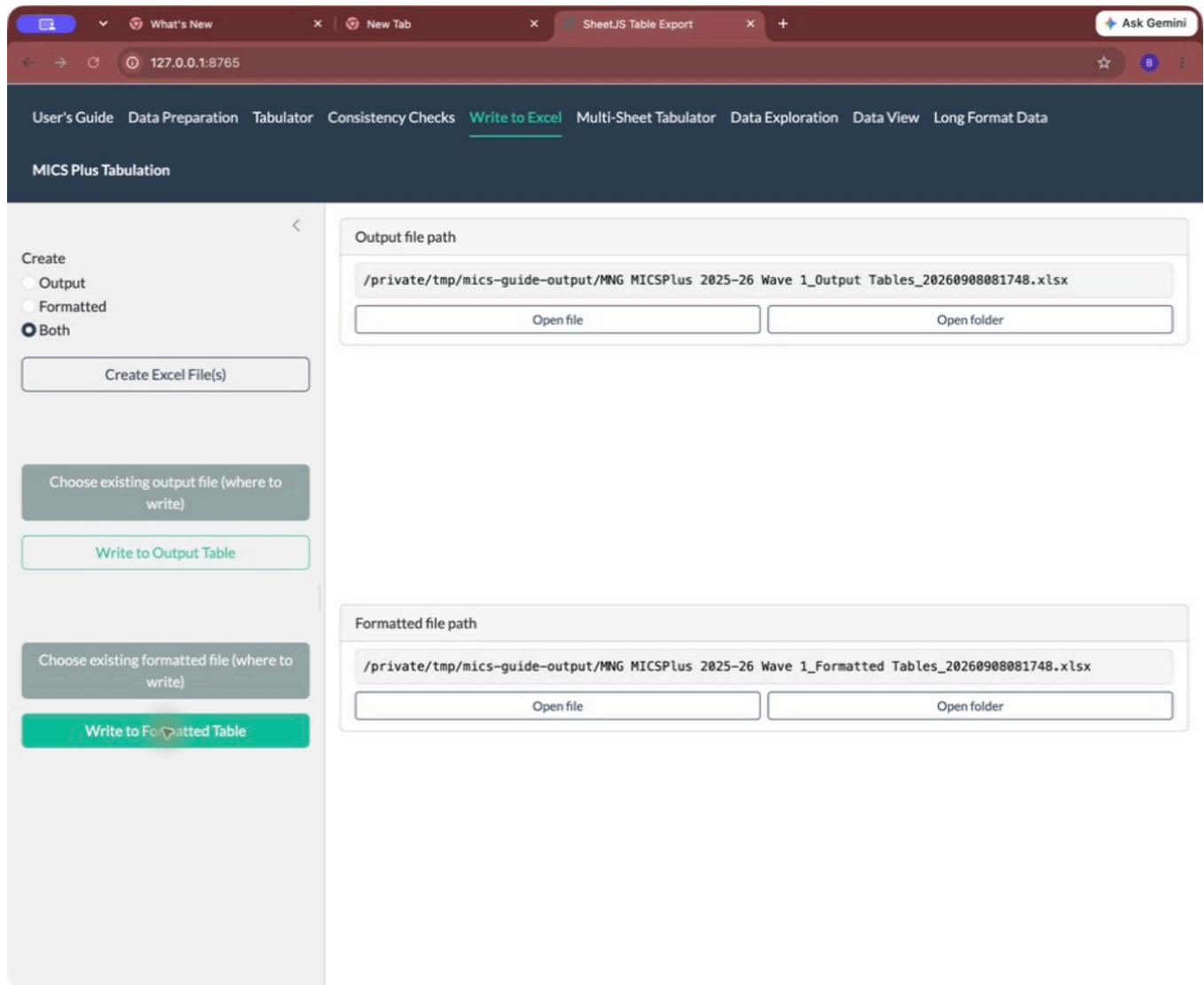


Figure 5: Write to Excel: create Both workbooks, write the current table with the two write buttons, and find the saved files using Open folder.

1. Click **Write to Excel**. Under **Create**, choose **Both** unless your survey lead requests one type only.
2. Click **Create Excel File(s)** once for this run. Two file paths appear on the right. This creates workbook copies; it does not fill every table.
3. Click **Write to Output Table**. Wait for the write to finish.
4. Click **Write to Formatted Table**. Wait again. Both buttons write the **current table** to its matching sheet.
5. Under **Formatted file path**, click **Open folder** to find the saved file, or **Open file** to view it. Check the expected sheet in Excel.

Output keeps numeric values for review. **Formatted** contains the table's presentation values and applicable notes. Close the workbook in Excel before writing another table. If **Open folder** does not work, copy the displayed folder path into File Explorer or Finder.

Next table: repeat **Tabulator > Run, Consistency Checks > Run Checks**, then the two **Write** buttons. Reuse the same workbooks; do not click **Create Excel File(s)** again for every table.

7. Run several tables together (optional)

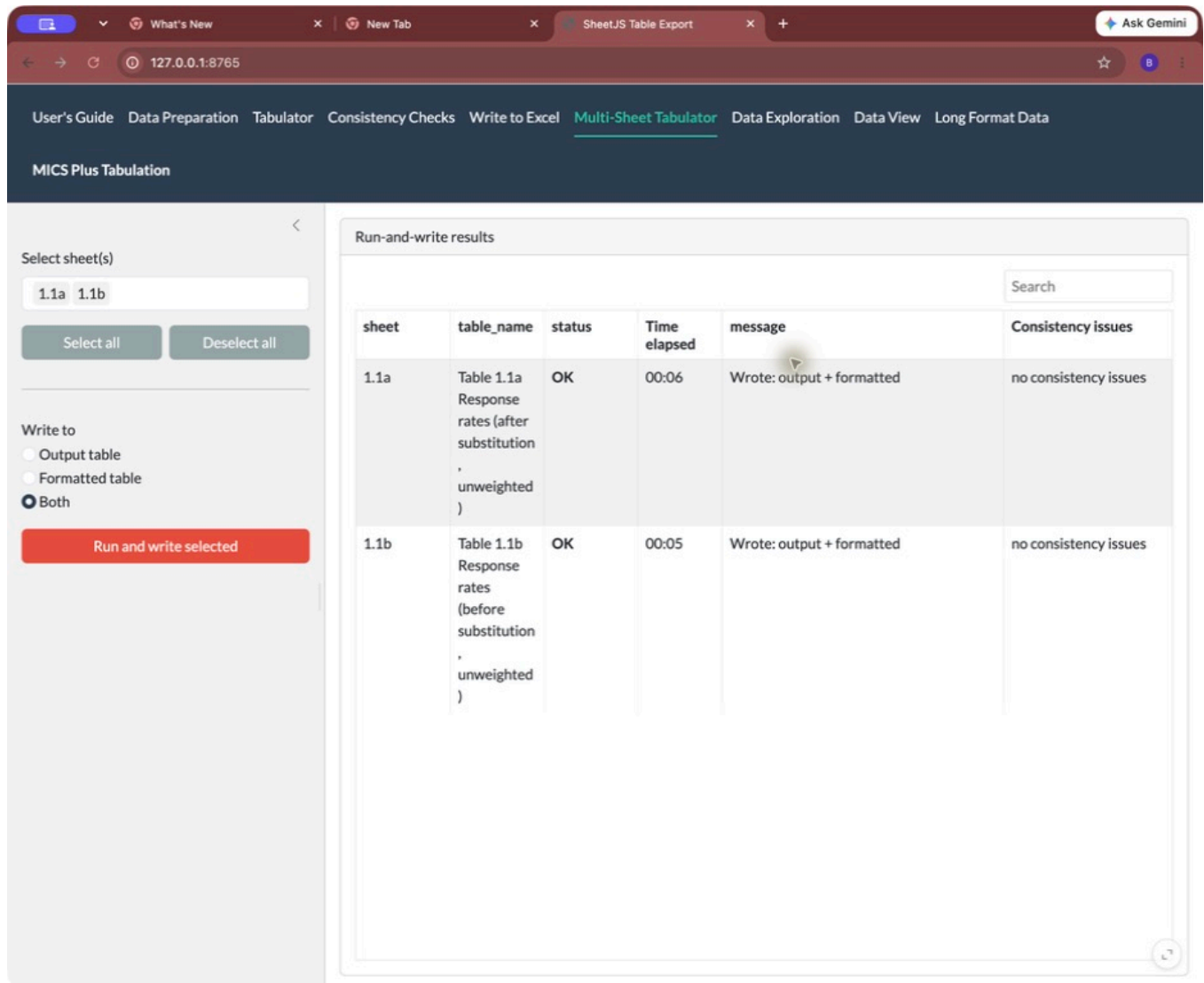


Figure 6: Multi-Sheet Tabulator: choose sheets, choose Both, then run and inspect one result row per sheet.

Use this after you can run and review one table successfully.

1. First create your workbook files in **Write to Excel** (step 6), if you have not already done so for this run.
2. Click **Multi-Sheet Tabulator**. Click **Select sheet(s)** and choose a table. Click again to add another. Use **Deselect all** to clear the selection.
3. Under **Write to**, choose **Both** or your agreed destination.
4. Click **Run and write selected** once. Keep your R environment and the browser open until it finishes.
5. When the completion message appears, click **OK**. Read every row in **Run-and-write results**, especially **status**, **message**, and **Consistency issues**. Use the horizontal scrollbar if a column is off the right edge.

Ready to continue: every requested table has completed and its results have been reviewed. This action writes tables as it runs; it does not wait for your review of each table. A completed write is not approval of a flagged result.

If a table fails or has consistency issues, note its code and message. Review it with your survey lead before sharing. Start with a few tables; use **Select all** when ready for the full run.

8. Download a data file (optional)

The screenshot displays the MICS Plus Tabulation application interface. On the left, the 'Long Format Data' section is active, featuring a 'Run all tables and prepare...' button and a 'Download output' section with buttons for Excel (.xlsx), CSV (.csv), and RDS (.rds). The main area shows an 'Output Table' with the following columns: STRUCTURE, STRUCTURE_ID, ACTION, FREQ, REF_AREA, SURVEY_ID, and TABL. The table contains 7 rows of data, all with the same values: STRUCTURE is 'dataflow', STRUCTURE_ID is 'MICS_PLUS_TABLE_CELL', ACTION is 'I', FREQ is 'I', REF_AREA is 'MNG', SURVEY_ID is 'MNG_MICSPLUS_2 025-26_W1', and TABL is '1.1a'. At the bottom of the table, it indicates '1-100 of 18162 rows' and a 'Show 100' dropdown menu.

Figure 7: Long Format Data: prepare the data with the button on the left, then use a download button below it.

Use this only if your survey lead asks for the tables as a combined data file. For ordinary presentation tables, use **Write to Excel** instead.

1. Complete **Data Preparation** first.
2. Click **Long Format Data**.
3. Click **Run all tables and prepare....** Wait while the application calculates the tables. This may take longer than running one table.
4. When the data preview appears without an error, find **Download output** in the left panel. Click **Excel (.xlsx)** for a spreadsheet, or the format your recipient requested.
5. Find the file in your browser's **Downloads** list or your computer's **Downloads** folder. Some browsers ask you to choose a save location.

What you receive: one combined file of calculated table cells. It is a different layout from the **Formatted** workbook. **CSV (.csv)** is a plain text table; **RDS (.rds)** is for someone working in R.

The download contains all prepared rows, including rows hidden by preview filters or pages. Check the country, period, and wave in the filename before sending it. If preparation reports a failed table, ask your survey lead to resolve the error and rerun before relying on a download.

9. Look at prepared data (optional)

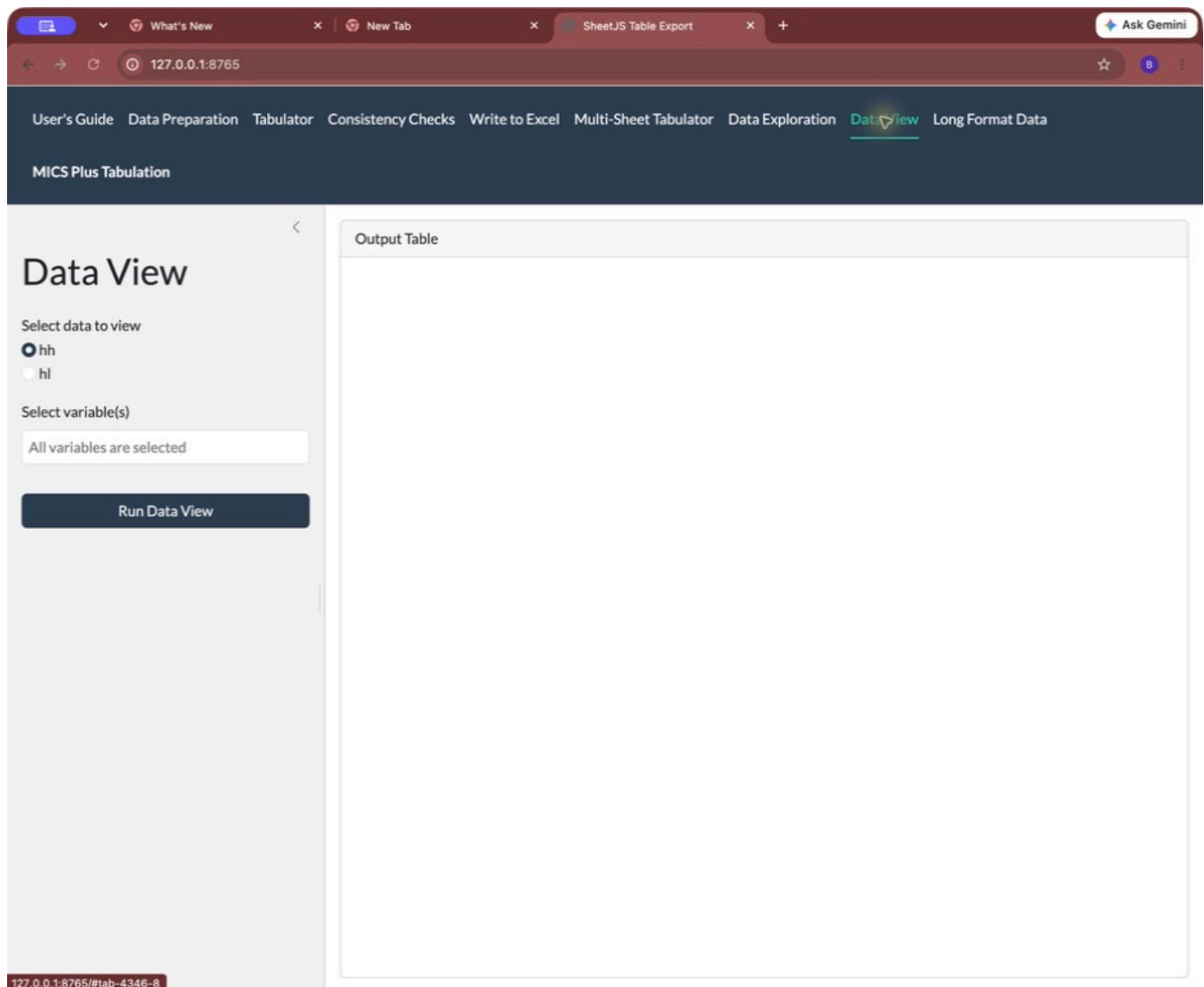


Figure 8: Data View: select the household or household members data, select variables, then click Run Data View.

This tool shows the data after preparation. Use it when your survey lead asks you to check a particular item. A **variable** is a named item in the data, like age or region.

1. Complete **Data Preparation**, then click **Data View**.
2. Under **Select data to view**, choose **hh** for households or **hl** for household members, as directed by your reviewer.
3. Select the variable names your reviewer gives you.
4. Click **Run Data View**. Read the preview and its labels.

Check: you are looking at the intended dataset and variables. This screen may show individual survey records. Share only what your survey team's data handling rules allow, and avoid including records in a help screenshot.

You can return to **Tabulator** by clicking its tab at the top. Viewing data here does not save a presentation table to Excel. The picture shows the controls before any individual records are displayed.

10. Explore a question (optional)

Figure 9: Data Exploration: choose the dataset, variables, and analysis type before clicking Run Data Explorer.

Use this for a question your survey lead wants to investigate. It does not replace the approved tabulation plan.

1. After **Data Preparation**, click **Data Exploration**.
2. Under **Select data set**, choose **hh** for households or **hl** for household members. Under **Select variable(s)**, choose the names your reviewer gives you.
3. Choose **Unweighted** or **Weighted**, as your reviewer specifies. If weighted, enter their exact variable name in **Type weight variable**. Do not guess it.
4. Choose **Frequency** to count categories of one variable, or **Cross table** to compare two variables. Select the second variable if requested.
5. Leave any filter box blank unless your reviewer supplies the exact text to use. Click **Run Data Explorer** and read the result.

Check: the selected data, variables, and weighting match the question you were asked to explore. These choices can change the meaning of a result. Ask your reviewer before treating it as a final survey finding.

11. Finish, return later, or get help

Finish your work

1. Write every reviewed table you want to keep to Excel (step 6).
2. Find the saved workbook. Open it and check the country, wave, and expected sheets. Keep your review notes with it.
3. Close the workbook, then close the application's browser tab.
4. In RStudio, click the red **Stop** button in the Console to stop the app. In another editor, use its R session interrupt/stop control. You may then close the editor. Saving an R workspace is not a substitute for saving the Excel workbooks.

When you return: reopen the same survey project, run `micsPlusTableR::run_app()`, and select your inputs again. The app does not automatically restore the previous browser session. For a fresh run, create new workbook files. To continue an existing workbook, ask your survey lead to help select it using the **Choose existing...** buttons in **Write to Excel**. Writing the same sheet again replaces its previously written values.

If something goes wrong

What you see	What to do next
Installation reports <code>setwd(startdir)</code> or a 00LOCK folder	Follow Installation cannot finish with your support person.
Download list unavailable	Check internet access and that the files repository is public, then click Load / refresh surveys .
Download says files already exist	Choose another download folder, or deliberately enable Replace existing files .
Missing file or wrong filename	Return to Data Preparation , select the correct file, and click Set again. Rerun and recheck the table.
Country or wave mismatch	Compare your dropdown choices with the files supplied by your team. Ask for a matched set if unsure.
Preparation error or missing variable	Record the message and send it to your survey lead. Do not edit the preparation code yourself.
No calculated table in Tabulator	Check that Set completed, choose a sheet, and click Run .
Workbook preview alone is blank or says Preview failed	Check your connection and whether cdn.jsdelivr.net is blocked. A cached library is not guaranteed; see workbook preview and caching . You can inspect the plan in Excel and continue tabulating. Save your results before restarting the app to retry.
Red consistency tab	Record the table code and the displayed differences. Have your survey lead review them before sharing.
Excel file cannot be written	Close that workbook in Excel, then try its Write button again.
Cannot find the saved workbook	Use Write to Excel > Open folder or the path shown there. For a data download, check Downloads instead.
Browser disconnected or refreshed	Save any work still available. Restart the app if needed and select the inputs again; check existing files before rewriting.

When asking for help, include: country and wave, table code, the button you clicked, and the exact error text. A screenshot of the message helps; leave individual survey records out of it. Your support person can check the installed package version in RStudio.

Read a calculation error

Calculation errors now identify the worksheet and, where available, the filter block, Excel row and column, statistic, and condition that failed. For example:

```
Worksheet 'Example', horizontal block 2, starting column 4 ...
Excel row 9, column 4; column condition: age >= 18; statistic: p
object 'age' not found
```

Here, **age** is missing from the prepared data used for that block. The row and column refer to positions in the Excel plan. A block is a group of columns that share a filter. An index alone (such as “In index: 1”) does not explain the cause; keep the complete message, including the expression and the final cause.

- **Prepared data source unavailable:** confirm that data preparation finished and that the plan names the correct household or household-member file.
- **Weight column not found or not numeric:** ask the survey lead to check the plan’s “weight by” entry and the variable created by the preparation script.
- **Unknown variable or invalid expression:** send the named condition and its Excel position to the survey lead to check spelling and R expression syntax.
- **Unsupported statistic:** the statistic cell contains a code that this direction of tabulation does not support. Ask the survey lead to correct it.
- **No data-source condition in B4:B7:** check that the selected worksheet is a table plan and contains its source/filter instructions in the expected cells.
- **Extra-table dimensions differ:** the preparation script’s supplied table must match the plan’s row count, with one label column and the value columns.
- **Comparison row or column counts differ:** align the two tables and check skipped header rows before comparing again.

After correcting the input, repeat data preparation if the script changed, then calculate and check the table again.

Installation cannot finish

Recognize the messages: Error in `setwd(startdir)` : cannot change working directory, or failed to lock directory followed by a folder such as `00LOCK-micsPlusTableR`. Ask your support person to help with these steps.

What happened: the first message means R's installer could not return to its starting folder. If that also prevents recovery, an unfinished installation can leave a lock folder that blocks later attempts. A lock error can also mean another installation is still running. These messages alone do not show why the starting folder became unavailable or prove a fault in the application.

Recover the installation

1. Save your work and the complete Console error. Close the app and other R sessions after any active installations finish.
2. Open your survey's **.Rproj** file from its normal, existing project folder. Keep that folder available throughout installation. Do not install from inside R's installed `micsPlusTableR` folder.
3. **Only if a lock error remains:** have your support person check the library path printed in the error and confirm no installer is running. Move only the reported stale lock folder to a separate backup location outside the library. It may contain the previous package needed for recovery. Do not delete the library or remove locks belonging to active installations.
4. Retry the original command in the Console:

```
remotes::install_github("barisr/micsPlusTableR")
```

5. After installation reports DONE (`micsPlusTableR`), verify that it loads:

```
library(micsPlusTableR)
packageVersion("micsPlusTableR")
```

No manual `setwd()` command or special installation options are required for this recovery route. If it fails again, keep the new error and stop repeating the installation until support has investigated it.

Prevent repeated working-directory failures

For support staff: run installation in a separate R process with an explicit temporary working directory outside the package library. Keep that directory alive until installation and cleanup finish, and pass the intended library paths to the process. This avoids relying on the user's current folder without changing their session's working directory. It requires a separate installation launcher; the standard GitHub command does not acquire this behavior from a change to `micsPlusTableR` alone.

Keep R's staged installation, loading checks, and locking enabled. Clearing a stale lock repairs a failed attempt; it does not fix the original cause. Maintainers should require successful installation and reinstallation checks before release and report a reproducible installer failure to R or `remotes`. This addresses the known failure mode, not every possible installation error.

For investigation, record `getwd()`, `.libPaths()`, and `sessionInfo()` before retrying. These commands only report information. Review paths for personal details before sharing the report. See the [R installation documentation](#).

Other editors and code systems (optional)

The RStudio route is sufficient for the whole workflow. If you already use another environment, the same package installation and launch commands apply. These are alternative ways to start R, not separate versions of the app.

1. **Create or open a survey folder** using your editor's folder/workspace control. An `.Rproj` file is RStudio's project shortcut; other editors do not need it just to run R. You can open the same folder in another editor.
2. **Start an R session in that folder.** In Positron, select an R interpreter and use its Console. In VS Code, configure the R extension and use **R: Create R terminal**. In Cursor, use compatible R tooling or run R in its terminal if R is available there. In Jupyter, choose an R kernel; a Python kernel cannot run these commands. R's own console also works.
3. **Check the folder in R** with `getwd()` before installing or launching. Opening an editor folder does not always change an existing R session's working directory. Start a fresh R session in your survey folder if needed.
4. **Install the package** using the two commands in **1D**, in that R session. Then check `here::here()` after installation: it shows the project root used for the default Excel output. If it selects a parent folder, supply your intended output directory explicitly when launching.
5. **Launch in a browser** with the command below, then follow sections 2-6 for uploads, preparation, table calculation, review, and Excel output:

```
micsPlusTableR::run_app(launch.browser = TRUE)
```

To choose the results folder explicitly, replace the example path below with an existing location on your computer. On Windows, use forward slashes, for example `C:/Users/YourName/Documents/MySurvey/results`:

```
micsPlusTableR::run_app(
  launch.browser = TRUE,
  output_dir = "/full/path/to/MySurvey/results"
)
```

Keep R running: the session or notebook cell stays busy while the app is open. Use its interrupt/stop control after saving your results. A terminal running R can usually be interrupted with **Ctrl+C**.

Remote or hosted environments: RStudio Server, Posit Workbench, and remote editor or notebook sessions need server-side package installation, access to the inputs, and a browser connection arranged by the administrator (often a proxy or port forwarding). Their output paths are on the server. The local `127.0.0.1` browser instructions assume R runs on your own computer; ask your support person for the hosted application's address and file-retrieval steps.